

Module 2 — PHP to Laravel

Rebuild your Project 1 REST API using Laravel.

Use PHP 8.4+ and Laravel 13.0+

- Read the `laravel_install.pdf` to understand the tools and scripts for the installation.
- (Optional) Read `lavavel_version_issue.pdf` if you want to understand why the version constraints.

What Changes?

Module 1: You wrote	Module 2: You use
Request routing and handlers	Routes and Controllers
SQL database operations	Eloquent Models
Database setup SQL	Migrations and seeders
Token checking	Sanctum and middleware

Laravel reduces repeated setup and request-handling work.

What Stays the Same?

- Your project's resources and API features.
- HTTP methods, requests, responses, and status codes.
- MySQL persistence and protected endpoints.
- Tests that check both successful and failed requests.

Compare behavior as you port; document any contract changes.

Module 2 Goal

1. Trace a request through Route → Controller → Model.
2. Port your Project 1 API to Laravel.
3. Verify database operations and Bearer Token protection.

Eloquent simplifies database work; SQL knowledge still matters.

Basics and Advanced

- The Basic directory covers the homework and project.
- For more detail, read the Advanced directory (optional).

Getting Started

1. Install Laravel following the `laravel_install.pdf` (same directory).
2. For HW4, run `run1-5.sh` with `student-api1-5/` in a separate directory.
3. Start the server and check an API response.

Windows: work in WSL2/Ubuntu. macOS: use Terminal.

Scripts make setup more consistent; check their prerequisites.

HW4 Reading Map

This overview introduces the goals. Continue with:

- `1_Laravel/Basics/` : framework, installation, routes, models, REST APIs.
- `2_ORM/Basics/` : ORM, Eloquent queries, and migrations.
- Lesson 5 code: migrations define tables; seeders add sample data.

These folders are under `module2/pdf/` (code under `module2/code/`).

For HW4, explain how these parts support your Project 2 API.